Description

Image





Caption

1. Polypropylene samples showing texture and transparency. © Chris Lefteri 2. Polypropylene glasses. © Thinkstock

The material

Polypropylene, PP, first produced commercially in 1958, is the younger brother of polyethylene - a very similar molecule with similar price, processing methods and application. Like PE it is produced in very large quantities (more than 30 million tons per year in 2000), growing at nearly 10% per year, and like PE its molecule-lengths and side-branches can be tailored by clever catalysis, giving precise control of impact strength, and of the properties that influence molding and drawing. In its pure form polypropylene is flammable and degrades in sunlight. Fire retardants make it slow to burn and stabilizers give it extreme stability, both to UV radiation and to fresh and salt water and most aqueous solutions.

Compositional summary

(CH2-CH(CH3))n

General properties

Density		895	-	909	kg/m^3
Price	*	0.898	-	0.974	GBP/kg
Date first used		1957			

Mechanical properties

Young's modulus		0.824	-	1.02	GPa
Shear modulus	*	0.327	-	0.36	GPa
Bulk modulus	*	0.872	-	0.961	GPa
Poisson's ratio	*	0.422	-	0.465	
Yield strength (elastic limit)		24.1	-	28.4	MPa
Tensile strength	*	26	-	50	MPa
Compressive strength	*	23.8	-	25	MPa

Values marked * are estimates.
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Elongation		112	-	483	% strain
Hardness - Vickers		8			HV
Fatigue strength at 10^7 cycles	*	7.64	-	8.02	MPa
Fracture toughness	*	1.18	-	1.31	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.0348	-	0.0365	

Thermal properties

Melting point	*	140	-	150	°C
Glass temperature		-24.2	-	-16.2	°C
Maximum service temperature	*	66.9	-	83.9	°C
Minimum service temperature	*	-25.2	-	-10.2	°C
Thermal conductor or insulator?		Good insulator			
Thermal conductivity	*	0.192	-	0.199	W/m.°C
Specific heat capacity		1.66e3	-	1.7e3	J/kg.°C
Thermal expansion coefficient		93.3	-	114	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity		9.95e21	-	2.17e24	µohm.cm
Dielectric constant (relative permittivity)		2.16	-	2.24	
Dissipation factor (dielectric loss tangent)		1.96e-4	-	2.04e-4	
Dielectric strength (dielectric breakdown)		17.6	-	18.4	MV/m

Optical properties

Transparency		Translucent			
Refractive index		1.48	-	1.5	

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		1	-	2	
Moldability		4	-	5	
Machinability		3	-	4	
Weldability		5			

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Excellent			

Soils, alkaline (clay)	Excellent
Wine	Excellent

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Excellent
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Excellent
Hydrofluoric acid (40%)	Excellent
Nitric acid (10%)	Excellent
Nitric acid (70%)	Excellent
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Excellent
Sulfuric acid (10%)	Excellent
Sulfuric acid (70%)	Excellent

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Limited use
Carbon tetrachloride	Limited use
Chloroform	Limited use
Crude oil	Acceptable
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Unacceptable
Vegetable oils (general)	Acceptable
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Excellent

Acetone	Excellent
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Unacceptable
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Unacceptable
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Poor

Durability: flammability

Flammability	Highly flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Acceptable
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	6.19e7			tonne/yr
Reserves, principal component	5.7e8	-	6.3e8	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 2.77	-	3.06	kg/kg
Embodied energy (virgin grade production)	* 65.9	-	72.6	MJ/kg
Water usage	* 37.2	-	41.2	l/kg

Material processing: energy

Polymer extrusion, embodied energy	* 8.14	-	9	MJ/kg
Polymer molding (injection, blow), embodied energy	* 23.1	-	26.3	MJ/kg
Machining, coarse, embodied energy	* 16.2	-	17.9	MJ/kg
Machining, fine, embodied energy	* 19.3	-	21.4	MJ/kg
Machining, grinding, embodied energy	* 22.4	-	25	MJ/kg

Material processing: climate change

Polymer extrusion, climate change (CO2-eq)	* 0.536	-	0.595	kg/kg
Polymer molding (injection, blow), climate change (CO2-eq)	* 1.63	-	1.87	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.06	-	1.17	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.26	-	1.4	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.46	-	1.64	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.914	-	1.01	kg/kg
Embodied energy, recycling	* 12.5	-	13.8	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 44	-	46.2	MJ/kg
Combustion CO2	* 3.06	-	3.22	kg/kg
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

PP is exceptionally inert and easy to recycle, and can be incinerated to recover the energy it contains. PP, like PE and PVC, is made by processes that are relatively energy-efficient, making them the least energy-intensive of commodity polymers. Its utility per kilogram far exceeds that of gasoline or fuel-oil (and its energy is stored and still accessible), so that production from oil will not disadvantage it in the near future

Recycle mark



Supporting information

Design guidelines

Standard grade PP is inexpensive, light and ductile but it has low strength. It is more rigid than PE and can be used at higher temperatures. The properties of PP are similar to those of HDPE but it is stiffer and melts at a higher temperature (165 - 170 C). Stiffness and strength can be improved further by reinforcing with glass, chalk or talc. When drawn to fiber PP has exceptional strength and resilience; this, together with its resistance to water, makes it attractive for ropes and fabric. It is more easily molded than PE, has good transparency and can accept a wider, more vivid range of colors. PP is commonly produced as sheet, moldings fibers or it can be foamed. Advances in catalysis promise new co-polymers of PP with more attractive combinations of toughness, stability and ease of processing. Mono-filaments fibers have high abrasion resistance and are almost twice as strong as PE fibers. Multi-filament yarn or rope does not absorb water, will float on water and dyes easily.

Technical notes

The many different grades of polypropylene fall into three basic groups: homopolymers (polypropylene, with a range of molecular weights and thus properties), co-polymers (made by co-Polymerization of propylene with other olefins such as ethylene, butylene or styrene) and composites (polypropylene reinforced with mica, talc, glass powder or fibers) that are stiffer and better able to resist heat than simple polypropylenes.

Typical uses

Ropes, automobile air ducting, parcel shelving and air-cleaners, garden furniture, washing machine tank, wet-cell battery cases, pipes and pipe fittings, beer bottle crates, chair shells, capacitor dielectrics, cable insulation, kitchen kettles, car bumpers, shatter proof glasses, crates, suitcases, artificial turf, thermal underwear.

Tradenames

Adpro, Amoco, Appryl, Aqualoy, Astryn, Cefor, Comalloy, Comshield, Dypro, EA36NA, Eltex P, Empee, Escorene, Ferrex, Ferrolene, Fortilene, Fotilene, Hifax, Hostalen PP, Latene, Marlex, Moplen, Multi-Flam, Multi-Pro, Nortuff, Novalen, Novolen, Nyloy, Petrothene, Polyfort, Polypro, Precolor, Pro Fax, Propak, Rexflex, Stamylyn, Starlylen, Statoil, Technoprene, Thermocomp, Vestolen, WPP, Washpen

Links

ProcessUniverse

Producers

Reference

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